HeuiChan (Terrence) Lim, Ph.D.

Assistant Professor of Computer Science

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Research Interests: Compilers, Dynamic Program Analysis, Software Engineering, Software Automation

ACADEMIC APPOINTMENTS

2024–Present	**Assistant Professor of Computer Science** Davidson College, Davidson, NC

EDUCATION

2024	**Ph.D. in Computer Science** The University of Arizona
2022	**M.S. in Computer Science** The University of Arizona
2018	**B.S. in Computer Information Systems** California State University, Chico

PUBLICATIONS AND PREPRINTS

Tianye Dai, Hammurabi Mendes, and **HeuiChan Lim**. "An LLVM-Based Optimization Pipeline for SPDZ." *Manuscript, arXiv*, 2025.

Kyra Dalbo, Yumna Ahmed, and **HeuiChan Lim**. "JITScope: Interactive Visualization of JIT Compiler IR Transformations." *Manuscript, arXiv*, 2025.

HeuiChan Lim. "Witness Test Program Generation through AST Node Combinations." In *IEEE International Conference on Software Testing, Verification and Validation Workshops (ICSTW)*, Naples, Italy, 2025.

HeuiChan Lim and Saumya Debray. "Directed Test Program Generation for JIT Compiler Bug Localization." *Manuscript, arXiv*, 2023.

HeuiChan Lim and Saumya Debray. "Automatically Localizing Dynamic Code Generation Bugs in JIT Compiler Back-End." In *Proceedings of the 32nd ACM SIGPLAN International Conference on Compiler Construction (CC)*, Montréal, QC, Canada, 2023.

HeuiChan Lim, Xiyu Kang, and Saumya Debray. "Modeling Code Manipulation in JIT Compilers." In *Proceedings of the 11th ACM SIGPLAN International Workshop on the State Of the Art in Program Analysis (SOAP)*, pp. 9–15, 2022.

HeuiChan Lim and Stephen Kobourov. "Visualizing JIT Compiler Graphs." In *29th International Symposium on Graph Drawing and Network Visualization (GD)*, 2021.

HeuiChan Lim and Saumya Debray. "Automated Bug Localization in JIT Compilers." In *Proceedings of the 17th ACM SIGPLAN/SIGOPS International Conference on Virtual Execution Environments (VEE)*, 2021.

A. Pyarelal, M. Valenzuela-Escárcega, R. Sharp, P. Hein, J. Stephens, P. Bhandari, **H. Lim**, S. Debray, and C. Morrison. "AutoMATES: Automated Model Assembly from Text, Equations, and Software." *Modeling and Managing Complicated Systems*, 2019.

GRANTS

2026	**Sweetnswitz**, $20,000 External research grant
2026	**Davidson Research Initiative (DRI)**, $5,250 Davidson College internal research grant

2026	**Faculty Study and Research (FS&R)**, $5,000 Davidson College internal research grant
2025	**Faculty Study and Research (FS&R)**, $5,000 Davidson College internal research grant

TEACHING

2024–Present	**Davidson College** CSC 221: Data Structures CSC 312: Software Design CSC 355: Compiler Design CSC 356: Computer Security
Fall 2023	**The University of Arizona** Graduate Teaching Assistant, CSC 436: Software Engineering *Supervisor: Ravi Sethi*
Fall 2018	**The University of Arizona** Graduate Teaching Assistant, CSC 460: Database Design *Supervisor: Lester McCann*
2017–2018	**California State University, Chico** Undergraduate Teaching Lab Assistant, CSC 311: Algorithms and Data Structures *Supervisor: Judy Challinger*
2016–2018	**California State University, Chico** Undergraduate Teaching Assistant, BADM 495: Business Dynamics, Applied Strategic Decision-Making *Supervisor: Hyunjung Kim*

RESEARCH EXPERIENCE

2024–Present	**Davidson College** Research on automated end-to-end debugging pipelines targeting AOT and JIT compilers, automated software systems, and software correctness, including AI-enhanced systems and their reliability.
2020–2023	**Graduate Research Assistant** The University of Arizona *Advisor: Saumya Debray* Dynamic program analysis of Just-in-Time compiler execution behavior for automated fault localization.
2019–2020	**Graduate Research Assistant** The University of Arizona *Advisor: Saumya Debray* Developed a Fortran-to-Python translator that extracts code and data from Fortran programs and generates equivalent Python programs.

PRESENTATIONS

2025	**HeuiChan Lim.** "Automated Dynamic Code Generation Bug Localization for Just-in-Time Compilers." Invited Guest Speaker, California State University, Chico.

ACADEMIC SERVICE

2026	**Program Committee Member** IEEE Annual Computer Security Applications Conference (ACSAC)
2026	**Program Committee Member** ACM SIGCSE Technical Symposium, Computing Education Research Track

2026 **Program Committee Member**
ACM SIGCSE Technical Symposium, Student Research Competition

2025 **Program Committee Member**
ACM SIGPLAN International Workshop on the State Of the Art in Program Analysis (SOAP)

STUDENT MENTORSHIP

2026 **Davidson Research Initiative (DRI)**

- Witness Generation for JIT Compiler Fault Localization.
- Test Code Quality Analysis.

2025-2026 **Honors Thesis**

- Honors Thesis advisor of Asuna Dai.
- Second reader of Kenan Wood's Honors Thesis.
- Committee member of Andrew Jones, DuBose Tuller, and Chadi Bsila's Honors Thesis.

2025–Present **Major Advisor**

- Seven undergraduate advisees.

STUDENT POSTERS

2026 N. Aleksii, R. Votta, Y. Yolcu, M. Fajardo. AssemblyViz: A Web-Based Assembly Visualizer for HYMN and RISC-V. Poster. Verna Miller Case Research and Creative Works Symposium, Davidson College, Davidson, NC. May 7, 2026.

2026 K. Atas, E. Jerjees, T. Makoni, H.B. Yavuzkara. Retrieval-Augmented Generation for Institutional Immigration Guidance. Poster. Verna Miller Case Research and Creative Works Symposium, Davidson College, Davidson, NC. May 7, 2026.

2026 R. Doctor, B. Chung, F. Howden, M. Mostafa. Developments of New Testing Methodologies for the Computer Science Department. Poster. Verna Miller Case Research and Creative Works Symposium, Davidson College, Davidson, NC. May 7, 2026.

OPEN-SOURCE SOFTWARE

2025 **Node Combination and Classification for AST Transformation (NCCAT)**
Published under the MIT License

2023 **Just-in-Time Compiler Intermediate Representation Modeler (JITCIRModeler)**
Published under the MIT License

2021 **Just-in-Time Compiler IR Visualization (JITCompilerIRViz)**
Published under the MIT License

PROFESSIONAL DEVELOPMENT

2026 **Teaching Open Source (TOS) Kits and Projects Workshop**
Worcester, MA
Participated in a faculty development workshop focused on integrating Teaching Open Source projects into software engineering education. Explored practical approaches and resources for incorporating open-source project work and engaging students in authentic software development practices.

2025 **Professors' Open Source Software Experience (POSSE) 2025 Workshop**
Norfolk, VA
Participated in a faculty development workshop focused on integrating open-source contributions into undergraduate computing education. Engaged with pedagogical strategies and practical tools for mentoring student participation in real-world open-source projects.

AWARDS AND HONORS

2017	**Jeffrey J. Hagen Memorial Scholarship** California State University, Chico
2017	**UPE Academic Achievement Award** California State University, Chico
2016	**Jeffrey J. Hagen Memorial Scholarship** California State University, Chico
2016	**First Place, Public Speech Rookie Tournament** California State University, Chico Informative Speech: *"Why Programming is Important for College Students"*
2015	**2015/2016 Andrew Southard Scholarship** California State University, Chico